Numerical Analysis Assignment 2

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Assessed exercise 2.1 Consider the following functions from $\mathbb{R}$ to $\mathbb{R}$. Prove that 0:

- is a simple root of the identity (also known as linear unit) $x \mapsto x =: \mathrm{id}_{\mathbb{R}}(x)$.
- is not a simple root of $x \mapsto x^2$.
- is a simple root of $x \mapsto |x|$.
- is a simple root of $x \mapsto \max\{x, 2x\}$.
- is not a simple root of the so-called rectified linear unit (ReLU) $x \mapsto \max\{0, x\}$.

SOLUTION:

1. To prove that 0 is a simple root of the identity function $f(x) = x$ we can observe that:
$$\phi(x) = \frac{\|f(x)\|}{\|x - 0\|} = \frac{\|x\|}{\|x\|} = 1 > 0 \quad \text{for } x \neq 0.$$
Hence,
$$\liminf_{x \to 0} \phi(x) = \liminf_{x \to 0} 1 = 1 > 0$$
so 0 is a simple root of f. $\square$

2. To prove that 0 is not a simple root of $f(x) = x^2$ we can observe that:
$$\phi(x) = \frac{\|f(x)\|}{\|x - 0\|} = \frac{\|x^2\|}{\|x\|} = \|x\| \quad \text{for } x \neq 0.$$
Hence,
$$\liminf_{x \to 0} \phi(x) = \liminf_{x \to 0} \|x\| = 0.$$
so 0 is not a simple root of f. $\square$

3. To prove that 0 is a simple root of $f(x) = |x|$ we can observe that:
$$\phi(x) = \frac{\|f(x)\|}{\|x - 0\|} = \frac{\| \, |x| \, \|}{\|x\|} = \frac{\|x\|}{\|x\|} = 1 > 0 \quad \text{for } x \neq 0.$$
Hence,
$$\liminf_{x \to 0} \phi(x) = 1 > 0$$
so 0 is a simple root of f. $\square$

4. To prove that 0 is a simple root of $f(x) = \max\{x, 2x\}$ we can observe that:
$$\phi(x) = \frac{\|f(x)\|}{\|x - 0\|} = \frac{\|\max\{x, 2x\}\|}{\|x\|} = \begin{cases} \frac{\|2x\|}{\|x\|} = 2 & ; x > 0 \\ \frac{\|x\|}{\|x\|} = 1 & ; x < 0 \end{cases} > 0 \quad \text{for } x \neq 0.$$
Hence,
$$\lim_{x \to 0^+} \phi(x) = 2 \text{ and } \lim_{x \to 0^-} \phi(x) = 1 \implies \liminf_{x \to 0} \phi(x) = 1 > 0.$$
so 0 is a simple root of f. $\square$

5. To prove that 0 is not a simple root of $f(x) = \max\{0, x\}$ we can observe that:
$$\phi(x) := \frac{\|f(x)\|}{\|x - 0\|} = \frac{\|\max\{0, x\}\|}{\|x\|} = \begin{cases} \frac{\|x\|}{\|x\|} = 1 & ; x > 0 \\ \frac{\|0\|}{\|x\|} = 0 & ; x < 0 \end{cases} \quad \text{for } x \neq 0.$$
Hence,
$$\lim_{x \to 0^+} \phi(x) = 1 \text{ and } \lim_{x \to 0^-} \phi(x) = 0 \implies \liminf_{x \to 0} \phi(x) = 0.$$
so 0 is not a simple root of f. $\square$

Assessed exercise 2.2

- Check that the trapezoidal rule is not exact on polynomials of degree two.
- Check that the midpoint rule is exact on polynomials of degree less than or equal to 1.
- Check that the midpoint rule is not exact on polynomials of degree two.

SOLUTION:

1. To prove that the trapezoidal rule is not exact on polynomials of degree two, we observe that given:

$$\mathcal{T}_a^b[f] := (b-a)\frac{f(a)+f(b)}{2}$$

For polynomials of degree 2 on the interval $[a,b]$ it holds that:

$$f(x) \equiv x^2 \implies \mathcal{E}_{\mathcal{T}_a^b}[f] = \left[\int_a^b x^2\,dx\right] - \left[(b-a)\,\frac{b^2+a^2}{2}\right] = \frac{b^3-a^3}{3} - \frac{(b-a)(b^2+a^2)}{2}$$

Exactness would require:

$$\frac{b^3-a^3}{3} \equiv \frac{(b-a)(b^2+a^2)}{2}$$

Which clearly does not hold for all values of $a, b \in \mathbb{R}$. Hence the trapezoidal rule is not exact on polynomial functions P with $\deg P = 2$.

Note that this result aligns with Theorem 4.7 which gives us:

$$\left[\left|\int_a^b f(x)\,dx - \mathcal{T}_a^b[f]\right|\right]_{f\in P_2} \leq \frac{1}{12}\,\|f''\|_{L^\infty}\,(b-a)^3 = \frac{(b-a)^3}{6} \neq 0\ ;\ b \neq a.$$

2. To prove that the midpoint rule is exact on polynomials of degree less than or equal to 1, we observe that given:

$$\mathcal{M}_a^b[f] := (b-a)\,f\left(\frac{a+b}{2}\right)$$

We have

- For polynomials of degree 0 on the interval $[a,b]$ it holds that:

$$f(x) \equiv 1 \implies \mathcal{E}_{\mathcal{M}_a^b}[f] = \left[\int_a^b 1\,dx\right] - [(b-a)\cdot 1] = (b-a) - (b-a) = 0.$$

- For polynomials of degree 1 on the interval $[a,b]$ it holds that:

$$f(x) \equiv x \implies \mathcal{E}_{\mathcal{M}_a^b}[f] = \left[\int_a^b x\,dx\right] - \left[(b-a)\cdot\left(\frac{a+b}{2}\right)\right] = \frac{b^2-a^2}{2} - \frac{b^2-a^2}{2} = 0.$$

Hence, for polynomial functions with $\deg P \leq 1$, the midpoint rule is exact.

Note that this result aligns with Theorem 4.7 which gives us:

$$\left[\left|\int_a^b f(x)\,dx - \mathcal{M}_a^b[f]\right|\right]_{f\in P_1} \leq \frac{1}{24}\,\|f''\|_{L^\infty}\,(b-a)^3 = \frac{1}{24}\cdot 0\cdot(b-a)^3 = 0.$$

3. To prove that the midpoint rule is not exact on polynomials of degree two, we observe that given:

$$\mathcal{M}_a^b[f] := (b-a)\, f\left(\frac{a+b}{2}\right)$$

For polynomials of degree 2 on the interval $[a,b]$ it holds that:

$$f(x) \equiv x^2 \implies \mathcal{E}_{\mathcal{M}_a^b}[f] = \left[\int_a^b x^2\, dx\right] - \left[(b-a)\left(\frac{a+b}{2}\right)^2\right] = \frac{b^3-a^3}{3} - \frac{(b-a)(a+b)^2}{4}.$$

Exactness would require:

$$\frac{b^3-a^3}{3} \equiv \frac{(b-a)(a+b)^2}{4}.$$

Which clearly does not hold for all values of $a, b \in \mathbb{R}$. Hence the midpoint rule is not not exact on polynomial functions P with $\deg P = 2$.

Note that this result aligns with Theorem 4.7 which gives us:

$$\left[\left|\int_a^b f(x)\, dx - \mathcal{M}_a^b[f]\right|\right]_{f\in P_2} \leq \frac{1}{24}\,\|f''\|_{L^\infty}\,(b-a)^3 = \frac{(b-a)^3}{12} \neq 0\ ;\ b \neq a.$$

Assessed exercise 2.3 We saw in lectures that, by using a Lagrange interpolation based quadrature with $n+1$ points, we are able to exactly integrate polynomials up to order n. However it was seen that with the three points a, $\frac{a+b}{2}$, and b, that is $n=2$, it was possible to integrate a third order polynomial. This is because the points were conveniently chosen.

Here we will work out how to choose four points so that we can integrate fourth order polynomials. Assuming that the points are given by a, the symmetric points $\frac{a+b}{2} \pm r$, and b, where $r \in \left(0, \frac{b-a}{2}\right)$ is to be determined.

1. Calculate the Lagrange polynomials associated to this.
2. Given that the weights associated to the quadrature are given by

$$\omega_0 = \omega_3 = \frac{(b-a)^3 - 12r^2(b-a)}{6\left((b-a)^2 - 4r^2\right)}, \qquad \omega_1 = \omega_2 = \frac{(b-a)^3}{3\left((b-a)^2 - 4r^2\right)}, \tag{1}$$

show that the quadrature is exact on polynomials of order four or less when $r = \frac{b-a}{2\sqrt{5}}$.

3. Check if it is exact on fifth order polynomials; for simplicity, set $a=-1$ and $b=1$.

SOLUTION:

1. To calculate the Lagrange polynomials, we begin by denoting:

$$x_0 = a, \qquad x_1 = \frac{a+b}{2} - r, \qquad x_2 = \frac{a+b}{2} + r, \qquad x_3 = b.$$

Then we apply the formula to compute the Lagrange polynomials:

$$l_i(x) = \prod_{\substack{j=0 \\ j\neq i}}^{n} \frac{x-x_j}{x_i - x_j}.$$

Which yields:

$$\begin{aligned} l_0(x) = \prod_{j\neq 0} \frac{x-x_j}{x_0-x_j} &= \frac{x-\left(\frac{a+b}{2}-r\right)}{a-\left(\frac{a+b}{2}-r\right)} \cdot \frac{x-\left(\frac{a+b}{2}+r\right)}{a-\left(\frac{a+b}{2}+r\right)} \cdot \frac{x-b}{a-b} \\ &= \frac{x-\left(\frac{a+b}{2}-r\right)}{\left(\frac{a-b}{2}+r\right)} \cdot \frac{x-\left(\frac{a+b}{2}+r\right)}{\left(\frac{a-b}{2}-r\right)} \cdot \frac{x-b}{a-b} \\ &= \frac{\left(x-\left(\frac{a+b}{2}-r\right)\right)\left(x-\left(\frac{a+b}{2}+r\right)\right)(x-b)}{(a-b)\left(\left(\frac{a-b}{2}\right)^2 - r^2\right)} \end{aligned}$$

$$l_1(x) = \prod_{j\neq 1} \frac{x-x_j}{x_1-x_j} = \frac{x-a}{\left(\frac{a+b}{2}-r\right)-a} \cdot \frac{x-\left(\frac{a+b}{2}+r\right)}{\left(\frac{a+b}{2}-r\right)-\left(\frac{a+b}{2}+r\right)} \cdot \frac{x-b}{\left(\frac{a+b}{2}-r\right)-b}$$
$$= \frac{x-a}{\left(\frac{b-a}{2}-r\right)} \cdot \frac{x-\left(\frac{a+b}{2}+r\right)}{-2r} \cdot \frac{x-b}{\left(\frac{a-b}{2}-r\right)}$$
$$= \frac{(x-a)\left(x-\left(\frac{a+b}{2}+r\right)\right)(x-b)}{2r\left(\left(\frac{b-a}{2}\right)^2-r^2\right)}$$

$$l_2(x) = \prod_{j\neq 2} \frac{x-x_j}{x_2-x_j} = \frac{x-a}{\left(\frac{a+b}{2}+r\right)-a} \cdot \frac{x-\left(\frac{a+b}{2}-r\right)}{\left(\frac{a+b}{2}+r\right)-\left(\frac{a+b}{2}-r\right)} \cdot \frac{x-b}{\left(\frac{a+b}{2}+r\right)-b}$$
$$= \frac{x-a}{\left(\frac{b-a}{2}+r\right)} \cdot \frac{x-\left(\frac{a+b}{2}-r\right)}{2r} \cdot \frac{x-b}{\left(\frac{a-b}{2}+r\right)}$$
$$= -\frac{(x-a)\left(x-\left(\frac{a+b}{2}-r\right)\right)(x-b)}{2r\left(\left(\frac{b-a}{2}\right)^2-r^2\right)}$$

$$l_3(x) = \prod_{j\neq 3} \frac{x-x_j}{x_3-x_j} = \frac{x-a}{b-a} \cdot \frac{x-\left(\frac{a+b}{2}-r\right)}{b-\left(\frac{a+b}{2}-r\right)} \cdot \frac{x-\left(\frac{a+b}{2}+r\right)}{b-\left(\frac{a+b}{2}+r\right)}$$
$$= \frac{x-a}{b-a} \cdot \frac{x-\left(\frac{a+b}{2}-r\right)}{\left(\frac{b-a}{2}+r\right)} \cdot \frac{x-\left(\frac{a+b}{2}+r\right)}{\left(\frac{b-a}{2}-r\right)}$$
$$= \frac{(x-a)\left(x-\left(\frac{a+b}{2}-r\right)\right)\left(x-\left(\frac{a+b}{2}+r\right)\right)}{(b-a)\left(\left(\frac{b-a}{2}\right)^2-r^2\right)}$$

Note as one would expect, integrating our above obtained Lagrange Polynomials: $\int_a^b l_i(x)dx\,;\, i=0,1,2,3$ in WolframAlpha returns the provided values of ω_i :

$$\omega_0 = \omega_3 = \frac{(b-a)^3 - 12r^2(b-a)}{6\left((b-a)^2-4r^2\right)}, \qquad \omega_1 = \omega_2 = \frac{(b-a)^3}{3\left((b-a)^2-4r^2\right)},$$

2. To prove that the quadrature is exact on polynomials of order four or less with $r = \frac{b-a}{2\sqrt{5}}$ we need to show that for $k = 0, 1, 2, 3, 4$:

$$\int_a^b x^k\,dx = \int_a^b \sum_{i=0}^{3} x_i^k\, l_i(x)\,dx = \sum_{i=0}^{3} x_i^k \int_a^b l_i(x)\,dx = \sum_{i=0}^{3} \omega_i x_i^k = Q[f].$$

With

$$x_0 = a, \qquad x_1 = \frac{a+b}{2} - r, \qquad x_2 = \frac{a+b}{2} + r, \qquad x_3 = b$$

- For $k = 0$ we have:

$$\int_a^b 1\,dx = b-a, \qquad Q[1] = \omega_0+\omega_1+\omega_2+\omega_3 = 2\omega_0+2\omega_1.$$

Evaluating $\omega_0+\omega_1$ yields:

$$\omega_0+\omega_1 = \frac{(b-a)^3-12r^2(b-a)}{6\left((b-a)^2-4r^2\right)} + \frac{(b-a)^3}{3\left((b-a)^2-4r^2\right)}$$
$$= \frac{(b-a)^3-12r^2(b-a)+2(b-a)^3}{6\left((b-a)^2-4r^2\right)}$$
$$= \frac{3(b-a)^3-12r^2(b-a)}{6\left((b-a)^2-4r^2\right)} = \frac{(b-a)\left(3(b-a)^2-12r^2\right)}{6\left((b-a)^2-4r^2\right)}$$
$$= \frac{(b-a)\cdot 3\left((b-a)^2-4r^2\right)}{6\left((b-a)^2-4r^2\right)} = \frac{b-a}{2}.$$

Hence $Q[1] = 2(\omega_0+\omega_1) = b-a = \int_a^b 1\,dx$.

- For $k = 1$ we have:

$$\int_a^b x\,dx = \frac{b^2 - a^2}{2}.$$

$$\begin{aligned} Q[x] &= \omega_0 x_0 + \omega_1 x_1 + \omega_2 x_2 + \omega_3 x_3 \\ &= \omega_0 a + \omega_1 \left(\frac{a+b}{2} - r\right) + \omega_2 \left(\frac{a+b}{2} + r\right) + \omega_3 b \\ &= \omega_0 a + \omega_1 \left(\frac{a+b}{2} - r\right) + \omega_1 \left(\frac{a+b}{2} + r\right) + \omega_0 b \\ &= (\omega_0 + \omega_1)(a+b) = \underbrace{\left(\frac{b-a}{2}\right)}_{\text{As per } k=0} (a+b) = \frac{b^2 - a^2}{2} \end{aligned}$$

Hence $Q[x] = \omega_0 x_0 + \omega_1 x_1 + \omega_2 x_2 + \omega_3 x_3 = \frac{b^2-a^2}{2} = \int_a^b x\,dx$.

- For $k = 2$ we have:

$$\int_a^b x^2\,dx = \frac{b^3 - a^3}{3}.$$

$$\begin{aligned} Q[x^2] &= \omega_0 x_0^2 + \omega_1 x_1^2 + \omega_2 x_2^2 + \omega_3 x_3^2 \\ &= \omega_0 a^2 + \omega_1 \left(\frac{a+b}{2} - r\right)^2 + \omega_1 \left(\frac{a+b}{2} + r\right)^2 + \omega_0 b^2 \\ &= \omega_0 (a^2 + b^2) + \omega_1 \left(\left(\frac{a+b}{2} - r\right)^2 + \left(\frac{a+b}{2} + r\right)^2\right) \\ &= \omega_0 (a^2 + b^2) + \omega_1 \left(2\left(\frac{a+b}{2}\right)^2 + 2r^2\right) \\ &= \omega_0 (a^2 + b^2) + \omega_1 \left(\frac{(a+b)^2}{2} + 2r^2\right) \\ &= \omega_0 \left(\frac{(a+b)^2 + (b-a)^2}{2}\right) + \omega_1 \left(\frac{(a+b)^2}{2} + 2r^2\right) \\ &= \frac{(b-a)\left((b-a)^2 - 12r^2\right)}{6\left((b-a)^2 - 4r^2\right)} \cdot \frac{(a+b)^2 + (b-a)^2}{2} + \frac{(b-a)^3}{3\left((b-a)^2 - 4r^2\right)} \left(\frac{(a+b)^2}{2} + 2r^2\right) \\ &= \frac{(b-a)}{6\left((b-a)^2 - 4r^2\right)} \left(\frac{\left((b-a)^2 - 12r^2\right)\left((a+b)^2 + (b-a)^2\right)}{2} + 2(b-a)^2 \left(\frac{(a+b)^2}{2} + 2r^2\right)\right) \\ &= \frac{(b-a)}{12\left((b-a)^2 - 4r^2\right)} \left(\left((b-a)^2 - 12r^2\right)\left((a+b)^2 + (b-a)^2\right) + 2(b-a)^2\left((a+b)^2 + 4r^2\right)\right) \\ &= \frac{(b-a)}{12\left((b-a)^2 - 4r^2\right)} \begin{pmatrix} (b-a)^2(a+b)^2 + (b-a)^4 - 12r^2(a+b)^2 - 12r^2(b-a)^2 \\ + 2(b-a)^2(a+b)^2 + 8(b-a)^2 r^2 \end{pmatrix} \\ &= \frac{(b-a)}{12\left((b-a)^2 - 4r^2\right)} \left(3(b-a)^2(a+b)^2 + (b-a)^4 - 12r^2(a+b)^2 - 4(b-a)^2 r^2\right) \\ &= \frac{(b-a)}{12\left((b-a)^2 - 4r^2\right)} \left(\left((b-a)^2 - 4r^2\right)\left(3(a+b)^2 + (b-a)^2\right)\right) \\ &= \frac{(b-a)}{12}\left(3(a+b)^2 + (b-a)^2\right) = \frac{(b-a)}{12}\left(3(a^2 + 2ab + b^2) + a^2 - 2ab + b^2\right) \\ &= \frac{(b-a)}{12}\left(4a^2 + 4ab + 4b^2\right) = \frac{(b-a)}{3}\left(a^2 + ab + b^2\right) = \frac{b^3 - a^3}{3}. \end{aligned}$$

Hence $Q[x^2] = \omega_0 x_0^2 + \omega_1 x_1^2 + \omega_2 x_2^2 + \omega_3 x_3^2 = \frac{b^3-a^3}{3} = \int_a^b x^2\,dx$.

- For $k = 3$ we have:

$$\int_a^b x^3\,dx = \frac{b^4 - a^4}{4}.$$

$$
\begin{aligned}
Q[x^3] &= \omega_0 x_0^3 + \omega_1 x_1^3 + \omega_2 x_2^3 + \omega_3 x_3^3 \\
&= \omega_0 a^3 + \omega_1 \left(\frac{a+b}{2} - r\right)^3 + \omega_1 \left(\frac{a+b}{2} + r\right)^3 + \omega_0 b^3 \\
&= \omega_0(a^3+b^3) + \omega_1 \left(\left(\frac{a+b}{2} - r\right)^3 + \left(\frac{a+b}{2} + r\right)^3\right) \\
&= \omega_0(a^3+b^3) + \omega_1 \left(\begin{array}{l}\left(\frac{a+b}{2}\right)^3 - 3\left(\frac{a+b}{2}\right)^2 r + 3\left(\frac{a+b}{2}\right) r^2 - r^3 \\ \qquad + \left(\frac{a+b}{2}\right)^3 + 3\left(\frac{a+b}{2}\right)^2 r + 3\left(\frac{a+b}{2}\right) r^2 + r^3\end{array}\right) \\
&= \omega_0(a^3+b^3) + \omega_1 \left(2\left(\frac{a+b}{2}\right)^3 + (-3+3)\left(\frac{a+b}{2}\right)^2 r + 6\left(\frac{a+b}{2}\right) r^2 + (-1+1) r^3\right) \\
&= \omega_0(a^3+b^3) + \omega_1 \left(2\left(\frac{a+b}{2}\right)^3 + 6\left(\frac{a+b}{2}\right) r^2\right) \\
&= \omega_0(a^3+b^3) + \omega_1 \left(\frac{(a+b)^3}{4} + 3(a+b) r^2\right) \\
&= \frac{(b-a)\left((b-a)^2 - 12r^2\right)}{6\left((b-a)^2 - 4r^2\right)}(a^3+b^3) + \frac{(b-a)^3}{3\left((b-a)^2 - 4r^2\right)}\left(\frac{(a+b)^3}{4} + 3(a+b)r^2\right)
\end{aligned}
$$

Let $s = a+b,\ d = b-a,\ \implies a = \dfrac{s-d}{2},\ b = \dfrac{s+d}{2}$

$$
\begin{aligned}
&\implies \frac{d\left(d^2 - 12r^2\right)}{6\left(d^2 - 4r^2\right)}(a^3+b^3) + \frac{d^3}{3\left(d^2 - 4r^2\right)}\left(\frac{s^3}{4} + 3sr^2\right) \\
a^3 + b^3 &= \frac{(s-d)^3 + (s+d)^3}{8} = \frac{2s^3 + 6sd^2}{8} = \frac{s(s^2+3d^2)}{4} \\
&\implies \frac{d\left(d^2 - 12r^2\right)}{6\left(d^2 - 4r^2\right)} \cdot \frac{s\left(s^2 + 3d^2\right)}{4} + \frac{d^3}{3\left(d^2 - 4r^2\right)}\left(\frac{s^3}{4} + 3sr^2\right) \\
&= \frac{s}{24\left(d^2 - 4r^2\right)}\left[d\left(d^2 - 12r^2\right)(s^2+3d^2) + 8d^3\left(\frac{s^2}{4} + 3r^2\right)\right] \\
&= \frac{s}{24\left(d^2 - 4r^2\right)}\left[d^3 s^2 + 3d^5 - 12ds^2r^2 - 36d^3r^2 + 2d^3s^2 + 24d^3r^2\right] \\
&= \frac{s}{24\left(d^2 - 4r^2\right)}\left[3d^3 s^2 + 3d^5 - 12ds^2r^2 - 12d^3r^2\right] \\
&= \frac{s}{24\left(d^2 - 4r^2\right)} \cdot 3d\left[d^2 s^2 + d^4 - 4s^2r^2 - 4d^2r^2\right] \\
&= \frac{s}{24\left(d^2 - 4r^2\right)} \cdot 3d\left(s^2 + d^2\right)\left(d^2 - 4r^2\right) \\
&= \frac{sd\left(s^2 + d^2\right)}{8} = \frac{(a+b)(b-a)\left((a+b)^2 + (b-a)^2\right)}{8} \\
&= \frac{\left(b^2 - a^2\right)\left(a^2 + b^2\right)}{4} = \frac{b^4 - a^4}{4}.
\end{aligned}
$$

Hence $Q[x^3] = \omega_0 x_0^3 + \omega_1 x_1^3 + \omega_2 x_2^3 + \omega_3 x_3^3 = \frac{b^4 - a^4}{4} = \int_a^b x^3\,dx$.

- For $k = 4$ we have: $\displaystyle\int_a^b x^4\,dx = \frac{b^5 - a^5}{5}$.

$$
\begin{aligned}
Q[x^4] &= \omega_0 x_0^4 + \omega_1 x_1^4 + \omega_2 x_2^4 + \omega_3 x_3^4 \\
&= \omega_0 a^4 + \omega_1 \left(\frac{a+b}{2} - r\right)^4 + \omega_1 \left(\frac{a+b}{2} + r\right)^4 + \omega_0 b^4 \\
&= \omega_0(a^4+b^4) + \omega_1 \left[\left(\frac{a+b}{2} - r\right)^4 + \left(\frac{a+b}{2} + r\right)^4\right]
\end{aligned}
$$

$$= \omega_0(a^4+b^4) + \omega_1\left(2\left(\frac{a+b}{2}\right)^4 + 12\left(\frac{a+b}{2}\right)^2 r^2 + 2r^4\right)$$

$$= \omega_0(a^4+b^4) + \omega_1\left(\frac{(a+b)^4}{8} + 3(a+b)^2 r^2 + 2r^4\right)$$

$$= \frac{(b-a)\left((b-a)^2-12r^2\right)}{6\left((b-a)^2-4r^2\right)}\left(a^4+b^4\right) + \frac{(b-a)^3}{3\left((b-a)^2-4r^2\right)}\left(\frac{(a+b)^4}{8} + 3(a+b)^2r^2 + 2r^4\right)$$

Let $s = a+b,\ d = b-a, \implies a = \frac{s-d}{2},\ b = \frac{s+d}{2}$

$$a^4+b^4 = \frac{(s-d)^4+(s+d)^4}{16} = \frac{s^4+6s^2d^2+d^4}{8}$$

$$\implies \frac{1}{24(d^2-4r^2)}\left[\frac{1}{2}d(d^2-12r^2)(s^4+6s^2d^2+d^4) + d^3s^4 + 24d^3s^2r^2 + 16d^3r^4\right]$$

At this point note:

$$\frac{b^5-a^5}{5} = \frac{1}{5}\left[\left(\frac{s+d}{2}\right)^5 - \left(\frac{s-d}{2}\right)^5\right]$$

And: $(s \pm d)^5 = s^5 \pm 5s^4d + 10s^3d^2 \pm 10s^2d^3 + 5sd^4 \pm d^5$

$$\implies (s+d)^5 - (s-d)^5 = 2\left(5s^4d + 10s^2d^3 + d^5\right)$$

$$\implies \frac{b^5-a^5}{5} = \frac{1}{80}\left(5ds^4 + 10d^3s^2 + d^5\right)$$

By WolframAlpha our previous expression evaluates to:

$$s^2\left(\frac{d^3}{8} + \frac{ds^2}{16}\right) + d^3\left(\frac{d^2}{48} - \frac{r^2}{6}\right)$$

$$= \frac{1}{80}\left(5ds^4 + 10d^3s^2 + d^5\right) + \frac{d^3}{120}\left(d^2 - 20r^2\right)$$

$$= \frac{b^5-a^5}{5} + \frac{(b-a)^3}{120}\left((b-a)^2 - 20r^2\right)$$

Hence to obtain exactness on fourth order polynomials, we require :

$$(b-a)^2 = 20r^2 \implies r = \frac{b-a}{2\sqrt{5}}$$

I.e. given $r = \frac{b-a}{2\sqrt{5}}$, it holds that:

$$Q[x^4] = \omega_0x_0^4 + \omega_1x_1^4 + \omega_2x_2^4 + \omega_3x_3^4 = \frac{b^5-a^5}{5} = \int_a^b x^4\,dx$$

Since we have now shown that the quadrature is exact on the monomials $x^k; k = 0,1,2,3$ and, provided that $r = \frac{b-a}{2\sqrt{5}}$, also for $k = 4$. It therefore holds by the linearity of the integral operator that the quadrature is exact on all polynomials of order four or less. I.e.:

$$\text{Given } r = \frac{b-a}{2\sqrt{5}},\ a,b \in \mathbb{R}, f \in \mathcal{P}_4 \,:\, \mathcal{E}_{\mathcal{Q}_a^b}[f] = 0$$

3. To check if the quadrature is exact on fifth order polynomials with $a = -1$ and $b = 1$, we can repeat the checks conducted above for $k = 0,1,2,3,4$ to the case $k = 5$ with $a = -1$ and $b = 1$.

   For $k = 5$ we have:

$$\int_{-1}^{1} x^5\,dx = \frac{1-1}{6} = 0.$$

$$Q[x^5] = \omega_0x_0^5 + \omega_1x_1^5 + \omega_2x_2^5 + \omega_3x_3^5$$
$$= \omega_0a^5 + \omega_1\left(\frac{a+b}{2} - r\right)^5 + \omega_1\left(\frac{a+b}{2} + r\right)^5 + \omega_0b^5$$

$$
\begin{aligned}
&= \omega_0(1^5 + (-1)^5) + \omega_1\left[\left(\frac{-1+1}{2} - r\right)^5 + \left(\frac{-1+1}{2} + r\right)^5\right] \\
&= 0 + \omega_1\Big[(0-r)^5 + (0+r)^5\Big] \\
&= \omega_1\big[-r^5 + r^5\big] = 0
\end{aligned}
$$

Hence $Q[x^5] = \omega_0 x_0^5 + \omega_1 x_1^5 + \omega_2 x_2^5 + \omega_3 x_3^5 = \frac{1^6-(-1)^6}{6} = 0 = \int_{-1}^{1} x^5\,dx$.

Meaning the quadrature is indeed exact on fifth order polynomials. (Since we can rescale the given interval $[-1, 1]$ and achieve the result generally $\forall\, I \in \mathbb{R}$).
